CS 473 (Spring 2025) Yiming Qin (yimingq3@illinois.edu)

Problem 7.3

Solution:

Algorithm 1 Optimal Rectangle Redrawing using Minimum-Cost Flow

Require: A set of rectangles $R = \{r_1, r_2, \ldots, r_n\}$, each with height 1, integer coordinates, and cost a_i.
Ensure: A redrawing R' that minimizes $\sum_{i=1}^{n} a_i \cdot \text{width}(r'_i)$.
1: **for** each rectangle r_i in R **do**
2:     Add a vertex v_i to V.
3: Add source s and sink t to V.
4: **for** each rectangle r_i in the first row **do**
5:     Add edge (s, v_i) to E.
6: **for** each row in R except the last row **do**
7:     **for** each rectangle r_i in the row **do**
8:         **for** each rectangle r_j in the next row **do**
9:             **if** r_i and r_j share a side **then**
10:                 Add edge (v_i, v_j) to E.
11: **for** each rectangle r_i in the last row **do**
12:     Add edge (v_i, t) to E.
13: **for** each rectangle r_i **do**
14:     Set lower bound $\ell(v_i) = 1$.
15:     Set upper bound $c(v_i) = \text{width}(r_i)$.
16:     Set cost $\text{cost}(v_i) = a_i$.
17: Set demand d to the total width of $U(R)$.
18: Compute flow f minimizing $\sum_{i=1}^{n} a_i \cdot \text{width}(r'_i)$.
19: Redraw each rectangle with $\text{width}(r'_i)$.
20: **return** Redrawing S'.

Proof of correctness:

- Since we have conservation of flow at every vertex in $V - \{s, t\}$, the redrawing S' remains a rectangle.
- Each rectangle width remains within valid bounds, i.e., $1 \leq \text{width}(r'_i) \leq \text{width}(r_i)$
- Step $7 \sim 11$ guarantees that the new layout maintains the adjacency structure.

Creating the directed graph takes polynomial time, computing the minimum-cost flow takes polynomial time. Thus, the overall algorithm runs in polynomial time.